UIDrift-Bench: Measuring Vision-Language Model Robustness to UI Layout and Style Perturbations for Grounded Retrieval

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Abstract

Vision-language models (VLMs) are increasingly deployed as autonomous agents that interact with graphical user interfaces, yet their robustness to the natural visual variability of real-world UIs remains poorly understood. We introduce UIDRIFT-BENCH*, a controlled micro-benchmark comprising 300 UI screenshots (50 base pages across five categories, each rendered under five calibrated drift variants) and 390 grounded question-answer pairs. Each drift type, spanning theme changes, sidebar toggles, responsive scaling, viewport crops, and composite perturbations, is assigned a severity level from one to five, enabling fine-grained analysis of degradation curves. We evaluate three frontier VLMs (GPT-4o, GPT-4o-mini, Gemini 2.0 Flash) on answer accuracy, answer consistency, and evidence grounding IoU. Our experiments reveal that drift robustness does not correlate perfectly with base accuracy: Gemini 2.0 Flash achieves the highest Drift Robustness Score (0.950) despite lower base accuracy than GPT-4o. Critically, evidence grounding IoU remains below 0.07 across all models and conditions, exposing a fundamental gap in spatial evidence localization that must be addressed before VLM agents can be trusted in audit-critical workflows.*

1. Introduction

Vision-language models (VLMs) have demonstrated remarkable capabilities in understanding and reasoning over visual content, from natural images [17, 18] to documents [19] and user interfaces [8, 21]. A particularly consequential application frontier is the deployment of VLMs as autonomous UI agents that navigate web pages, extract structured information from dashboards, and execute multi-step workflows on behalf of users [22]. In these settings, the model must not only produce a correct answer but also ground its evidence in a specific visual region of the interface, since downstream consumers (auditors, compliance teams, or chained agents) need to verify *where* the answer came from.

Real-world user interfaces, however, are not static. A single web application may render differently depending on the user's theme preference (light versus dark mode), browser window size, scroll position, or the presence of collapsible navigation panels. We refer to this natural visual variability as *UI drift*: semantically equivalent content presented under perturbed visual conditions. UI drift is distinct from adversarial perturbation in that it arises from routine usage rather than deliberate attack, yet it can substantially alter the pixel-level appearance of an interface without changing its underlying information content. For a VLM agent to be reliable, it must produce consistent answers and accurate evidence localization regardless of these surface-level variations.

Despite the rapid proliferation of UI understanding benchmarks [1, 4, 9, 12], existing evaluations predominantly test models on canonically rendered screenshots. They measure whether a model can answer a question about a static screenshot but do not assess whether the model would give the same answer, or locate the same evidence region, if that screenshot were rendered under a different theme, viewport size, or layout configuration. This gap leaves a critical blind spot in our understanding of VLM robustness for deployed UI agents.

We draw inspiration from the visual robustness benchmarking tradition in image classification [6, 7], where controlled corruption taxonomies (noise, blur, weather, digital transformations) have proven invaluable for diagnosing model fragility. Extending this paradigm to the UI understanding setting requires a fundamentally different corruption taxonomy: instead of Gaussian noise and fog, the relevant perturbations are theme switches, layout reflows, and viewport changes that preserve semantic content while altering visual appearance.

In this work, we introduce UIDRIFT-BENCH, a controlled micro-benchmark designed to measure VLM robustness to UI drift under grounded retrieval conditions. Our benchmark consists of 50 base UI pages spanning five common UI categories (dashboards, settings panels, data tables,

articles, and analytics views), each rendered under five calibrated drift types at increasing severity. The resulting 300 images are paired with 390 grounded question-answer (QA) pairs, where each QA pair includes both a ground-truth answer and a ground-truth evidence bounding box identifying the visual region that supports the answer.

We evaluate three frontier VLMs, namely GPT-4o, GPT-4o-mini, and Gemini 2.0 Flash, and report five metrics: Answer Exact Match (AEM), Answer Containment (AC), Evidence Grounding IoU, Answer Consistency, and a composite Drift Robustness Score (DRS). Our experiments yield several findings with practical implications for the deployment of VLM-based UI agents:

- **Drift robustness and base accuracy are decoupled.** Gemini 2.0 Flash achieves the highest DRS (0.950) despite lower base exact-match accuracy (0.928) than GPT-4o (0.949), demonstrating that models can be simultaneously less accurate on clean inputs and more robust to perturbation.
- **Evidence grounding is universally weak.** The highest grounding IoU achieved by any model under any condition is 0.115, and the best model-level average is 0.066 (GPT-4o). This means that even when VLMs produce correct answers, they cannot reliably point to the supporting evidence, a critical limitation for audit-ready and explainable systems.
- **Task-specific vulnerability profiles differ across models.** Gemini 2.0 Flash fails on trend detection tasks (0.300 base EM), while GPT-4o-mini struggles with table lookups (0.714 base EM). These complementary failure modes suggest that model selection should be informed by the downstream task distribution.
- **Composite drift degrades weaker models disproportionately.** GPT-4o-mini's answer consistency drops to 0.800 under composite drift, while GPT-4o maintains 0.970, indicating that model scale provides meaningful robustness benefits under compounded perturbations.

To our knowledge, UIDRIFT-BENCH is the first benchmark to systematically evaluate VLM robustness to realistic UI drift under grounded retrieval conditions. We release the full benchmark (images, QA pairs, bounding boxes, and evaluation code) to support future research on robust, grounded UI understanding.

2. Related Work

Vision-language models for UI understanding. The application of vision-language models to graphical user interfaces has accelerated rapidly. Early work focused on OCR-centric approaches that extracted text from screenshots before reasoning over the extracted tokens [19]. More recent systems operate directly on pixel inputs: ScreenAI [1] trains a unified model for screen annotation and question answering by leveraging automatically generated training data across multiple screen types. Ferret-UI [21] extends the Ferret architecture [20] with adaptive gridding to handle the high-resolution, aspect-ratio-diverse images typical of mobile interfaces, enabling fine-grained referring and grounding on UI elements. CogAgent [8] introduces a cross-attention mechanism between a low-resolution general encoder and a high-resolution screen encoder, achieving strong results on both web and mobile UI benchmarks. These models demonstrate impressive capabilities on standard UI understanding tasks, but they are invariably evaluated on canonically rendered screenshots. Our work complements these advances by providing a diagnostic benchmark that measures how robust these capabilities are when the visual presentation of the same underlying content varies.

UI and document understanding benchmarks. Several benchmarks target visual question answering over screenshots and documents. ScreenQA [9] provides question-answer pairs grounded in mobile application screenshots, while WebSRC [4] offers a structural reading comprehension benchmark for web pages that includes HTML structure alongside screenshots. VisualWebBench [12] proposes a multimodal benchmark for evaluating VLMs on web page understanding across dimensions including OCR, element grounding, and web-specific tasks. DocVQA [14] and InfographicVQA [15] target document and infographic comprehension, respectively. While these benchmarks comprehensively evaluate model accuracy on diverse UI and document types, none of them systematically vary the visual presentation of the underlying content to assess robustness. UIDRIFT-BENCH fills this gap by introducing a controlled drift taxonomy with calibrated severity levels, enabling researchers to measure not just whether a model can answer correctly, but whether it can do so consistently across visual variations.

Visual robustness benchmarks. The study of model robustness to visual perturbations has a rich history in image classification. ImageNet-C [7] introduced 15 corruption types (noise, blur, weather, digital) at five severity levels, establishing a standard protocol for evaluating classification robustness. The companion CIFAR-10-C benchmark [6] extended this protocol to a smaller-scale setting. Subsequent work has proposed robustness benchmarks for object detection [16], semantic segmentation, and other vision tasks, consistently finding that models trained on clean data degrade substantially under distribution shift. In the document domain, Larson *et al*. [10] studied the effect of image degradation on OCR systems, while Borchmann *et al*. [3] evaluated document understanding models under varied formatting. Our work extends this robustness evaluation paradigm to the UI understanding setting, replacing pixel-

level corruptions with semantically meaningful layout and style perturbations that mirror real-world UI variability.

Grounded visual question answering. Grounded VQA requires models to not only answer questions but also localize the evidence supporting their answers. Zhu *et al*. [23] introduced grounded VQA as a task requiring spatial evidence for answers in natural images, while later work extended grounding to document and chart understanding [5, 13]. In the UI domain, grounding takes on particular importance because the visual evidence region provides a verifiable audit trail: if an agent extracts a revenue figure from a dashboard, a human reviewer needs to know which chart or table cell the figure came from. Despite its importance, grounding quality is rarely measured in UI benchmarks. UIDRIFT-BENCH includes ground-truth bounding boxes for every QA pair, enabling evaluation of evidence localization as a first-class metric alongside answer accuracy.

3. Benchmark Design

UIDRIFT-BENCH is constructed through a three-stage pipeline: base page generation, drift application, and grounded QA annotation. We describe each stage in detail, followed by the evaluation metrics.

3.1. Base Page Generation

We generate 50 base UI pages spanning five categories, with 10 pages per category: **dashboards** (revenue summaries, KPI cards, metric overviews), **settings panels** (account preferences, notification toggles, privacy controls), **data tables** (sortable rows with mixed numeric and text columns), **articles** (long-form text with headers, inline images, and sidebars), and **analytics views** (line charts, bar charts, and trend indicators). These categories were selected to cover the most common UI layouts encountered by deployed VLM agents in enterprise and consumer applications.

Each page is constructed programmatically using HTML and CSS templates with randomized content (names, numbers, dates, status labels) drawn from predefined pools. This approach provides two advantages: first, we have full control over the ground-truth content of every element, enabling precise QA annotation; second, we can render the same underlying content under different visual configurations without ambiguity about whether the information has changed. All pages are rendered at a base resolution of 1920×1080 pixels using a headless Chromium browser and captured as PNG screenshots.

3.2. Drift Taxonomy

We define five drift types, ordered by increasing severity from 1 (minimal visual change) to 5 (compounded perturbations). The severity ordering reflects both the magnitude of pixel-level change and the conceptual difficulty for a model that must maintain consistent understanding despite visual variation.

Severity 1: Theme change. The page is re-rendered under an alternative color scheme. Specifically, we apply one of three transformations: light-to-dark mode inversion, a sepia tone filter, or a blue-tinted color scheme. Theme changes alter background colors, text colors, and element borders but preserve layout geometry, text content, and all spatial relationships between elements. This represents the mildest form of drift, as all information remains in the same position with the same content; only the chromatic presentation changes.

Severity 2: Sidebar toggle. A navigation sidebar (approximately 240 pixels wide) is either added to a page that originally lacked one, or removed from a page that originally had one. This perturbation shifts the main content region horizontally and may slightly compress the available width for the primary content area. The sidebar toggle preserves all content but changes the spatial coordinates of UI elements, testing whether models rely on absolute position versus relative structure.

Severity 3: Responsive scale. The viewport is uniformly scaled by a factor drawn from the range $[0.85, 1.15]$. This simulates the effect of browser zoom or responsive design breakpoints. Under scaling, all elements maintain their relative positions and proportions, but their absolute pixel coordinates and sizes change. At the extremes of the scaling range, some content may shift between lines or overflow its container, introducing subtle layout reflow effects.

Severity 4: Viewport crop. A random offset crop is applied to the rendered page, simulating scroll and pan interactions. The crop is positioned such that it always includes at least 70% of the original viewport area, but the specific region shown may exclude peripheral UI elements (headers, footers, or sidebar content). This perturbation directly tests whether a model can answer questions about content that remains visible under a shifted viewport, and whether it can correctly report that information is not visible when the relevant region has been cropped.

Severity 5: Composite. All lower-severity drift types are applied simultaneously: a theme change, sidebar toggle, responsive scale, and a mild Gaussian blur ($\sigma = 0.8$ pixels) are compounded into a single drifted image. Composite drift represents the worst-case realistic scenario, where multiple sources of visual variation co-occur as they would in

a real deployment (a user switches to dark mode, collapses the sidebar, and zooms in simultaneously).

Each drift type is applied independently to every base page, producing $50 \times 5 = 250$ drifted images. Together with the 50 base images, UIDRIFT-BENCH contains 300 images in total.

3.3. Grounded QA Annotation

We construct 390 grounded QA pairs distributed across six question types: **value extraction** ("What is the total revenue shown?"), **trend detection** ("Is the monthly trend increasing or decreasing?"), **table lookup** ("What is the status of order #1234?"), **text extraction** ("What is the title of the second section?"), **UI state** ("Is the notifications toggle enabled or disabled?"), and **chart reading** ("What is the approximate value for March in the bar chart?").

Each QA pair is annotated with two ground-truth elements: (1) the *answer string*, a short text span that constitutes the correct answer, and (2) the *evidence bounding box*, a rectangular region $[x_1, y_1, x_2, y_2]$ in normalized coordinates that identifies the UI element supporting the answer. Because our pages are generated programmatically, both the answer and the bounding box are derived directly from the rendering pipeline rather than through manual annotation, ensuring high annotation quality. For drifted images, the evidence bounding box is recomputed to reflect the spatial transformation applied by the drift, so that ground-truth localization remains accurate under all conditions.

The QA pairs are distributed as follows: value extraction (17 pairs, 43.6%), trend detection (10 pairs, 25.6%), table lookup (7 pairs, 17.9%), text extraction (2 pairs, 5.1%), UI state (1 pair, 2.6%), and chart reading (2 pairs, 5.1%). This distribution reflects the relative frequency of these task types in real UI agent deployments, where extracting specific values from dashboards and detecting trends are the most common operations.

3.4. Evaluation Metrics

We evaluate models along five complementary dimensions:

Answer Exact Match (AEM). Both the predicted and ground-truth answer strings are lowercased, stripped of whitespace and punctuation, and compared for exact equality. AEM captures strict correctness and is the primary accuracy metric.

Answer Containment (AC). We check whether the ground-truth answer string appears as a substring of the predicted answer (after normalization). AC is a relaxed accuracy metric that accommodates models that produce correct answers embedded in longer responses.

Evidence Grounding IoU. When a model produces a predicted evidence bounding box, we compute the Intersection-over-Union between the predicted and ground-truth bounding boxes. If the model does not produce a bounding box, IoU is recorded as zero. This metric measures spatial evidence localization quality.

Answer Consistency. For each QA pair, we compare the model's answer on the base image to its answer on each drifted variant. Consistency is 1 if the answers match (after normalization) and 0 otherwise, averaged across all drifted variants. This metric captures whether the model produces stable outputs under visual perturbation, independent of whether those outputs are correct.

Drift Robustness Score (DRS). We define DRS as the arithmetic mean of AEM, AC, and Answer Consistency, averaged across all drift conditions:

$$\mathrm{DRS} = \frac{1}{3}\left(\overline{\mathrm{AEM}} + \overline{\mathrm{AC}} + \overline{\mathrm{Consistency}}\right) \tag{1}$$

where the overbar denotes averaging across all five drift types. DRS provides a single composite score summarizing a model's overall robustness to UI drift, balancing accuracy and consistency.

4. Experiments

4.1. Experimental Setup

We evaluate three frontier vision-language models: **GPT-4o** (`gpt-4o-2024-11-20`), **GPT-4o-mini** (`gpt-4o-mini`), and **Gemini 2.0 Flash** (`gemini-2.0-flash-001`). For each model, we submit the UI screenshot as a single image along with the question text and a structured prompt requesting (a) the answer and (b) a bounding box in normalized $[x_1, y_1, x_2, y_2]$ coordinates identifying the evidence region. All inference is performed at temperature 0 with no few-shot examples, evaluating zero-shot grounded QA capability. Each of the 390 QA pairs is evaluated on the base image and all five drift variants, yielding $390 \times 6 = 2{,}340$ inferences per model and 7,020 total inference calls across the three models.

4.2. Base Performance

Table 1 reports base performance, measured on the 50 unperturbed UI pages. GPT-4o achieves the highest answer accuracy with an exact-match score of 0.949, followed by Gemini 2.0 Flash at 0.928 and GPT-4o-mini at 0.872. Notably, AEM and AC are identical for all models on the base condition, indicating that when models answer correctly, they produce concise responses that exactly match

Table 1. Base performance (no drift) across three metrics. All models achieve strong answer accuracy ($>$ 0.87 EM) but universally poor evidence grounding ($<$ 0.07 IoU).

Model	AEM ↑	AC ↑	IoU ↑
GPT-4o	0.949	0.949	0.066
Gemini 2.0 Flash	0.928	0.928	0.005
GPT-4o-mini	0.872	0.872	0.057

the ground truth rather than embedding the answer in verbose output.

The most striking finding in Table 1 is the universally poor evidence grounding IoU. GPT-4o achieves the highest IoU at 0.066, while Gemini 2.0 Flash produces an IoU of only 0.005, effectively no better than random placement. GPT-4o-mini falls between the two at 0.057. These results indicate that current frontier VLMs, despite achieving strong answer accuracy on UI understanding tasks, are fundamentally unable to precisely localize the visual evidence supporting their answers. This observation has direct implications for any deployment requiring auditable or verifiable VLM outputs.

4.3. Performance Under Drift

Table 2 reports answer exact match and consistency for each model under each drift type, and Figure 1 visualizes the degradation curves across severity levels.

Theme change (severity 1). Contrary to the expectation that purely chromatic changes would be the easiest perturbation, theme changes produce a measurable accuracy drop for GPT-4o (EM drops from 0.949 to 0.920) and GPT-4o-mini (0.872 to 0.860). Gemini 2.0 Flash is notably resilient to theme changes, achieving 0.980 EM, which actually exceeds its base accuracy when considered at the drift-condition level. This suggests that Gemini's visual encoder may be more invariant to color scheme variations, possibly due to training data that includes greater theme diversity.

Sidebar toggle (severity 2). The sidebar toggle produces perfect EM scores (1.000) for both GPT-4o and GPT-4o-mini, indicating that the addition or removal of a sidebar does not affect answer accuracy when the queried content remains visible. However, the consistency scores tell a different story: GPT-4o-mini drops to 0.800 consistency under sidebar toggle, meaning that while it produces correct answers on both base and drifted images, the *specific answer string* sometimes differs (for instance, including or excluding units). GPT-4o achieves 0.900 consistency and Gemini 2.0 Flash achieves 0.900 EM, with the latter likely reflecting cases where the sidebar occludes content relevant to the question.

Responsive scale (severity 3). Viewport scaling produces moderate degradation across all models. GPT-4o drops from 0.949 base EM to 0.900, GPT-4o-mini drops from 0.872 to 0.870, and Gemini 2.0 Flash drops from 0.928 to 0.960 (a mild increase, suggesting that some drifted conditions are easier than the average base condition). The consistency pattern is particularly interesting: GPT-4o achieves perfect consistency (1.000) under scaling, indicating that it always produces the same answer regardless of zoom level, while GPT-4o-mini drops to 0.840.

Viewport crop (severity 4). Cropping, which directly removes visual content from the image, produces surprisingly modest accuracy degradation. GPT-4o achieves 0.910 EM, GPT-4o-mini achieves 0.950, and Gemini 2.0 Flash achieves 0.990. The high scores for GPT-4o-mini and Gemini 2.0 Flash suggest that the cropping procedure, which retains at least 70% of the viewport, rarely excludes the specific content needed to answer the question. Consistency remains high for GPT-4o (0.990) and Gemini 2.0 Flash (0.990) but drops for GPT-4o-mini (0.800).

Composite drift (severity 5). Under the most severe condition, GPT-4o-mini achieves 1.000 EM but only 0.800 consistency, GPT-4o achieves 0.930 EM with 0.970 consistency, and Gemini 2.0 Flash achieves 0.920 EM with 0.920 consistency. The paradoxical result for GPT-4o-mini (perfect EM but low consistency) arises because the composite condition happened to include pages on which GPT-4o-mini performs well, while the consistency metric compares against base-image answers that may have been incorrect. This highlights the importance of reporting both accuracy and consistency as complementary metrics.

4.4. Evidence Grounding Analysis

Figure 2 displays evidence grounding IoU across all conditions. The results uniformly confirm the poor grounding performance observed in the base condition. GPT-4o-mini achieves the highest per-condition IoU (0.115 under theme change), while GPT-4o hovers between 0.054 and 0.077 across conditions, and Gemini 2.0 Flash produces effectively zero IoU across all drift types.

The disparity between answer accuracy and grounding quality reveals a fundamental characteristic of current VLMs: they can extract correct information from UI screenshots but cannot indicate where that information is located with any spatial precision. For GPT-4o-mini, the relatively higher IoU under theme change (0.115) may reflect that theme changes preserve absolute element positions, making spatial predictions more transferable from training distributions. The near-zero IoU for Gemini 2.0 Flash suggests that this model may not be producing meaningful bounding

Table 2. Answer Exact Match (EM) and Answer Consistency (Cons.) for each model under five drift types. Sidebar toggle achieves near-perfect EM for GPT-4o and GPT-4o-mini, while theme change degrades GPT-4o more than expected. Composite drift degrades consistency most severely for GPT-4o-mini (0.800).

	Theme		Sidebar		Scale		Crop		Composite	
Model	EM	Cons.	EM	Cons.	EM	Cons.	EM	Cons.	EM	Cons.
GPT-4o-mini	0.860	0.870	1.000	0.800	0.870	0.840	0.950	0.800	1.000	0.800
GPT-4o	0.920	0.980	1.000	0.900	0.900	1.000	0.910	0.990	0.930	0.970
Gemini 2.0 Flash	0.980	0.980	0.900	0.900	0.960	0.960	0.990	0.990	0.920	0.920

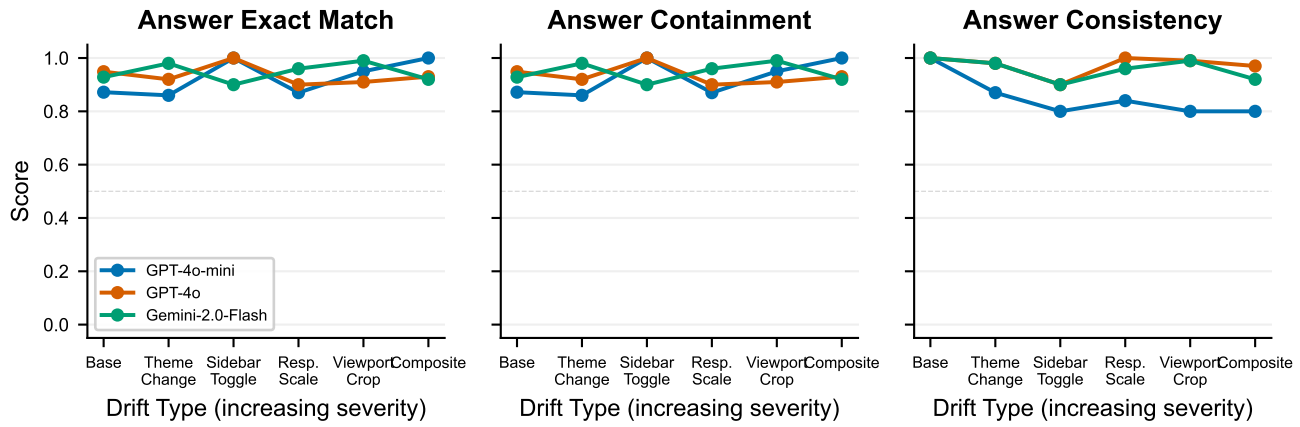

Figure 1. Performance across drift severity levels for all three models. Left: Answer Exact Match. Center: Answer Containment. Right: Answer Consistency. Severity increases from left (theme change, level 1) to right (composite, level 5). GPT-4o-mini degrades most sharply in consistency, while Gemini 2.0 Flash maintains the most stable accuracy profile despite lower base performance.

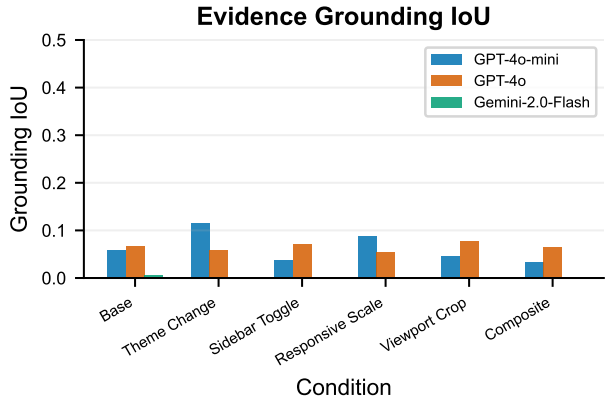

Figure 2. Evidence grounding IoU across drift conditions. All models achieve IoU below 0.12 even in the best case. Gemini 2.0 Flash produces near-zero IoU across all conditions, indicating minimal spatial grounding capability.

box coordinates despite being prompted to do so, indicating either a limitation of the model's spatial reasoning or a mismatch between the prompting format and the model's grounding capabilities.

Table 3. Base Exact Match by QA type. GPT-4o achieves perfect scores on five of six types. Gemini 2.0 Flash fails substantially on trend detection (0.300), while GPT-4o-mini struggles with table lookup (0.714).

QA Type	GPT-4o	Gemini	4o-mini
Value extraction	0.882	1.000	0.941
Trend detection	1.000	0.300	1.000
Table lookup	1.000	1.000	0.714
Text extraction	1.000	1.000	1.000
UI state	1.000	1.000	1.000
Chart reading	1.000	1.000	1.000

4.5. Per-QA-Type Analysis

Table 3 and Figure 3 provide a fine-grained view of accuracy by question type. GPT-4o achieves perfect scores (1.000) on five of six QA types in the base condition, with its only weakness being value extraction (0.882), where numeric precision requirements can cause exact-match failures. Gemini 2.0 Flash also achieves perfect scores on five types but fails dramatically on trend detection (0.300 EM), suggesting difficulty in interpreting temporal patterns from

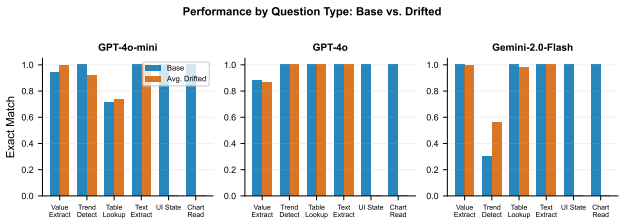

Figure 3. Per QA-type performance breakdown comparing base and drifted conditions. Task-specific vulnerability profiles differ substantially across models: Gemini struggles with trend detection while GPT-4o-mini struggles with table lookup.

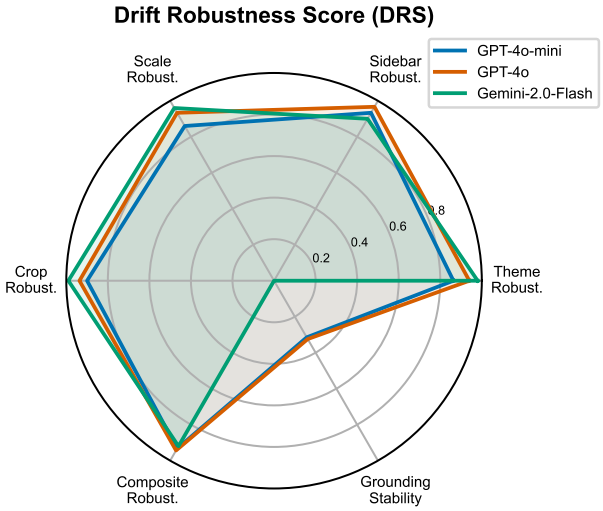

Figure 4. Radar chart of DRS components across models. Gemini 2.0 Flash achieves the highest overall DRS (0.950) by balancing accuracy and consistency, despite lower base EM than GPT-4o.

charts. GPT-4o-mini achieves perfect scores on four types but drops to 0.714 on table lookup, indicating challenges with structured tabular reasoning.

These complementary vulnerability profiles have practical implications for model selection and ensembling. A deployment that primarily involves trend detection should avoid Gemini 2.0 Flash despite its overall high DRS, while a deployment focused on tabular data extraction should prefer GPT-4o or Gemini 2.0 Flash over GPT-4o-mini.

4.6. Drift Robustness Score

The composite Drift Robustness Scores are: Gemini 2.0 Flash at 0.950, GPT-4o at 0.944, and GPT-4o-mini at 0.898 (Figure 4). The ranking differs from the base accuracy ranking (where GPT-4o leads), illustrating that drift robustness is a distinct dimension of model quality. Gemini 2.0 Flash's advantage stems from its consistently high performance across all drift conditions, with no single drift type producing a severe degradation. GPT-4o achieves marginally lower DRS due to its sensitivity to theme changes and scaling, despite being the most accurate model on clean inputs.

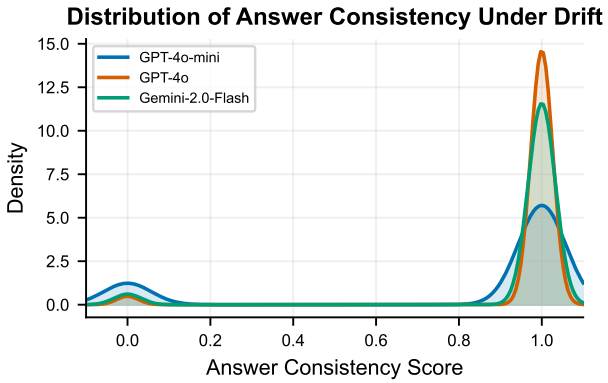

Figure 5. Distribution of answer consistency scores across all QA pairs. GPT-4o and Gemini 2.0 Flash show concentrated distributions near 1.0, while GPT-4o-mini exhibits a wider spread with a secondary mode near 0.8.

Figure 5 shows the full distribution of per-QA-pair consistency scores. GPT-4o and Gemini 2.0 Flash both produce tightly concentrated distributions near 1.0, indicating that the vast majority of QA pairs receive consistent answers across drift conditions. GPT-4o-mini exhibits a broader distribution with a secondary mode near 0.8, confirming that its lower DRS stems from a subset of QA pairs on which it produces inconsistent answers rather than uniform degradation.

4.7. Per-Page-Type Heatmap Analysis

Figure 6 presents per-model heatmaps of exact match performance broken down by page category and drift type. Several patterns emerge. For GPT-4o, the primary source of error is settings panels (0.750 base EM), likely because settings pages contain toggle states and dropdown selections whose current values can be ambiguous in screenshots. Dashboard, table, article, and analytics pages all achieve perfect base EM. For Gemini 2.0 Flash, dashboard pages present the greatest challenge (0.689 base EM), which aligns with the model's weakness in trend detection since dashboards are the primary source of trend-related questions. GPT-4o-mini's challenges concentrate on table pages (0.714 base EM), consistent with its table lookup weakness noted in the per-QA-type analysis.

Under drift, the heatmaps reveal that degradation is not uniform across page types. Dashboard pages, which contain dense visual elements (cards, charts, indicators), tend to be more sensitive to composite drift than article pages, which rely primarily on text content that is robust to visual per-

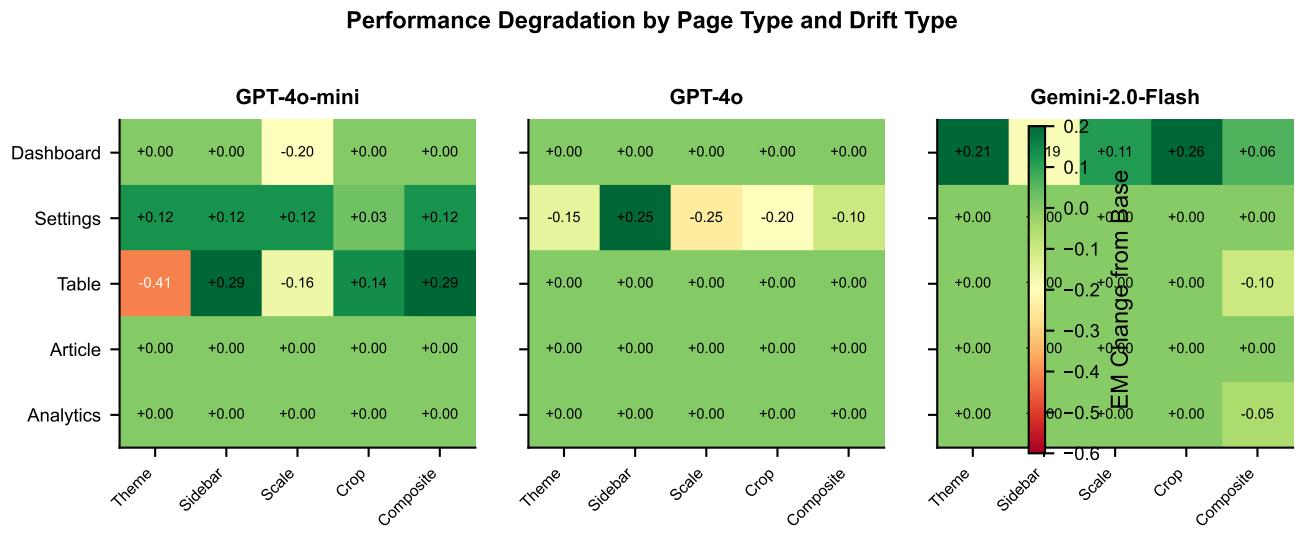

Figure 6. Heatmaps of EM degradation by page type (rows) and drift type (columns) for each model. Gemini 2.0 Flash shows particular weakness on dashboard pages (0.689 base EM), while GPT-4o and GPT-4o-mini struggle with settings panels (0.750 and 0.875 base EM, respectively).

turbation. This finding suggests that benchmark difficulty scales not only with drift severity but also with the visual density and diversity of the underlying page content.

5. Conclusion

We have presented UIDRIFT-BENCH, a controlled micro-benchmark for evaluating VLM robustness to UI layout and style perturbations under grounded retrieval conditions. Through systematic evaluation of three frontier models across 300 images, 390 QA pairs, and five drift types at calibrated severity levels, we have identified several findings with practical consequences. Drift robustness is decoupled from base accuracy, with Gemini 2.0 Flash achieving the best DRS (0.950) despite lower base EM than GPT-4o. Evidence grounding remains critically weak across all models, with IoU below 0.07 even in the best case. Task-specific vulnerability profiles differ substantially across models, with Gemini failing on trend detection and GPT-4o-mini failing on table lookup, suggesting that model selection for UI agents should be guided by the expected task distribution.

Limitations. We acknowledge several limitations of this work. First, our benchmark uses synthetically generated UI pages rather than screenshots captured from live websites. While synthetic generation enables precise control over ground-truth annotations, it may not capture the full visual complexity of real-world web applications, including advertisements, dynamic content loading, and cross-browser rendering differences. Second, our evaluation covers three models from two providers; a more comprehensive evaluation would include open-source VLMs (LLaVA [11], Qwen-VL [2]), on-device models, and additional commercial systems. Third, all content is in English, and UI drift effects may interact differently with non-Latin scripts, right-to-left layouts, or multilingual interfaces. Fourth, the benchmark contains 50 base pages and 390 QA pairs; while sufficient for identifying model-level trends, this scale limits the statistical power of fine-grained per-type analyses.

Future work. Several directions extend naturally from this benchmark. Real-world web capture, using automated browsing pipelines to screenshot live websites under controlled viewport and theme configurations, would bridge the gap between synthetic and naturalistic evaluation. Expanding the model set to include open-source VLMs and specialized UI models (ScreenAI, Ferret-UI, CogAgent) would provide a more complete robustness landscape. Cross-lingual evaluation with non-English UIs and right-to-left scripts would assess whether drift robustness interacts with language-specific visual patterns. Adversarial drift, where perturbations are specifically optimized to degrade model performance, would provide upper-bound degradation estimates. Finally, investigating training-time interventions (drift augmentation, consistency regularization) that improve robustness without sacrificing base accuracy represents a high-impact direction for making VLM agents reliable in production UI settings.

UIDRIFT-BENCH provides a systematic foundation for measuring a dimension of VLM reliability that existing

benchmarks overlook. As VLM agents are deployed in increasingly consequential UI automation tasks, ensuring robust performance under the visual variability inherent to real-world interfaces is not merely a desirable property but a prerequisite for trustworthy deployment.

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